

ThumbSwype: Thumb-to-Finger Gesture Based Text-Entry for Head Mounted Displays

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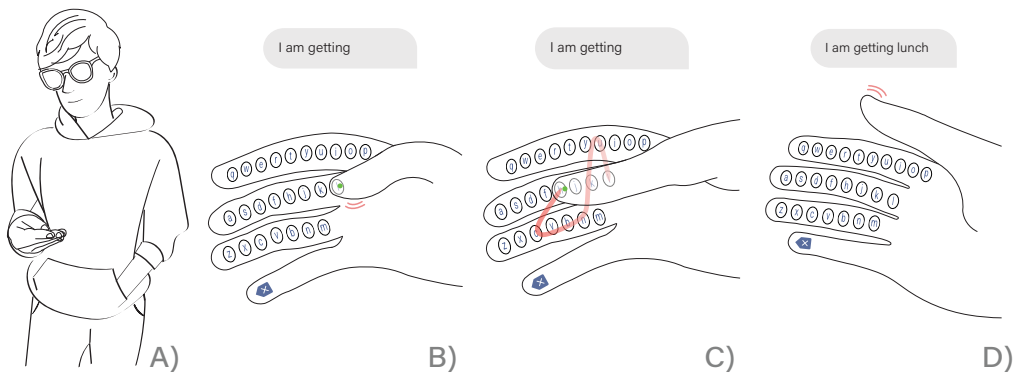


Fig. 1. Typing with ThumbSwype. a) The user wearing a head mounted display sees the keyboard overlaid on their fingers. b) A swipe gesture is initiated by touching down onto the relevant letter. c) Touch is maintained as the thumb moves comfortably over the remaining letters, similar to a smartphone swipe keyboard. d) The thumb is moved away from the keyboard, and the swiped word "lunch" is entered in the text box.

Designing a comfortable, familiar, and efficient one-handed text entry method for Head-Mounted Displays (HMDs) remains a significant challenge. Existing midair typing systems induce fatigue, while novel techniques often demand extensive training or sacrifice input efficiency. Consequently, we introduce **ThumbSwype**, a novel thumb-to-finger text entry technique that adapts smartphone swipe typing for HMDs. Users see the traditional QWERTY keyboard overlaid on their index, middle, and ring fingers, allowing them to perform swipe gestures with their thumb to type words. In an evaluation study ($N=16$), participants achieved a mean of 14.52 words per minute (WPM), which is 63.8% of their smartphone swipe-typing performance, with a peak

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average of 20.2 WPM. We compare ThumbSwipe's performance with related work, and discuss directions for future improvement.

CCS Concepts: • **Human-centered computing** → **Text input**; **Interaction design**.

Additional Key Words and Phrases: Hand Proximate User Interface, In-Situ Text Entry

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1 Introduction

Designing an efficient, intuitive and ergonomic one-handed text entry method remains an open challenge. One-handed techniques employing midair gestures [38, 50, 58, 66, 67], while promising, can lead to arm fatigue over extended use [35]. To address these limitations, researchers have explored alternative approaches [26, 29, 37, 84], including non-standard keyboard layouts that leverage thumb-to-finger interactions [24, 39, 42, 48, 49]. These methods significantly increase training time due to their unfamiliarity. Other solutions focus on mitigating situational impairments or encumbrance through micro-gesture interactions [27, 39, 53, 82]. However, they optimize for precision and stability while trading off input efficiency. This highlights the need for a unimanual text-entry method that combines comfort, familiarity and efficiency without the drawbacks of existing solutions.

With the aim of addressing this gap, we present ThumbSwipe: a novel, one-handed gesture-based text entry technique. ThumbSwipe works by overlaying the QWERTY keyboard layout on the user's fingers, and having the user perform thumb to finger swipes for quick and familiar smartphone analogous Word Gesture Keyboards (WGKs). Such gestures are also comfortable [14, 37] and socially acceptable [22]. The keyboard is visualized via the use of a deformable interface [20] following the posture of the hand, enabling users to identify the target locations for keys without having to rely on memory [23]. Additionally, it leverages self-bodied tactile feedback, allowing users to easily distinguish the start and end of gestures. Finally, other portions of the hand are utilized for additional text entry functionality, such as the radial surface of the index finger for selecting alternate suggestions and the little finger for deleting words.

To understand how intra-digit swipes work and improve the accuracy of the system, we first explored different noise filtering and shape matching algorithms for our technique. Results ($N=12$) show that a slower smooth One Euro Filter [9] and the Dynamic Time Warping Dependent [65] shape matching algorithm combination performed the best with 88.76% top 5 word prediction accuracy. To further validate the technique, an empirical study ($N=16$) evaluates ThumbSwipe, with users capable of averaging 14.52 WPM. The average highest phrase being entered at 20.2 WPM, nearly equivalent to that of their average smartphone input speeds at 22.75 WPM. Users additionally reported a mean comfort rating of 5.43 on 7 and 76.3 on the System Usability Scale [52]. We further analyze the correlations between smartphone to ThumbSwipe performance, and discuss ways of improving the overall system for future work.

Our main contributions are as follows:

- Developing ThumbSwipe, a novel, unimanual gesture based text entry technique for HMDs leveraging existing smartphone typing familiarity.
- Analysis of gestures performed in a three-dimensional deformable space, with a focus on comparing noise filtering and shape matching algorithms.
- Evaluating ThumbSwipe with novice and competent smartphone swipe users resulting in an average of 14.52 WPM, 63.8% of their smartphone swipe typing average.

2 Related Work

2.1 One-Handed Text Entry

A multitude of text-entry techniques exist for one-handed midair input, making use of WGKs [29, 58, 66, 67], or performing character-level inputs [38, 50]. As these methods utilize hand movements in free space they can lead to arm fatigue during prolonged use [35] and may lack reliability in constrained environments. Furthermore, the lack of haptic feedback [18, 58] in midair systems can hinder their effectiveness for eyes-free operation, as users lose the physical confirmation typically associated with keyboard typing. Other studies investigate novel techniques for text entry, such as character drawing [37], wrist movements [26, 29], and new keyboard layouts [24, 39, 42, 48, 49]. These approaches demonstrate potential in reducing fatigue associated with midair gestures, but face challenges on account of unfamiliarity, either due to novelty of the layout, novelty of the interaction technique, or some combination of both. Approaches aimed at in-situ computing or situational impairments often rely on micro-gesture interactions [17, 27, 28, 39, 48, 53, 82]. Some designs [17, 28] rely on an external surface for input. Other methods, aiming for robustness in constrained settings, often reduce the effective input space, compromising efficiency and achieving only 5–13 WPM [48, 53, 81, 82]. It's worth noting nearly all the techniques discussed here treat the hand itself purely as an input space, and expect users to perform an in-direct mental mapping by visualizing the keyboard on a separate medium such as an external monitor, or a floating window on HMDs.

For unencumbered text input, eye-gaze techniques [45, 54] offer hands-free interaction but are slow (9–11 WPM) and contribute to eye fatigue and longer task times [21]. Speech-to-text enables high input speed (150 WPM) and fast error correction [64], but raises privacy issues [19], limiting social usability. Discrete thumb-to-finger input, akin to smartphone use, may offer better social acceptability without physical encumbrance.

ThumbSwype aims to improve on these criteria by using ergonomic thumb-to-finger interactions, preserving the familiar QWERTY layout, and expanding the interaction space to three fingers for better efficiency and input precision. We additionally visualize the keyboard directly onto the surface of the hand itself, which stands to be more intuitive than in-direct mapping schemes [20].

2.2 Thumb-to-Finger Interactions

Thumb-to-finger interactions represent a viable ergonomic medium of input. Earlier studies demonstrate that users can reliably differentiate between 16 on-finger buttons [37], while subsequent research has expanded on these interactions, validating the functional workspaces of hand-based interfaces [10, 15, 20, 71]. Studies show that thumb-to-finger interactions are not only precise but also comfortable, highlighting their practical viability for implementing one-handed interactions [15, 23, 63]. Such microgestures are particularly advantageous due to their discrete nature, which contributes to social acceptability [33, 34, 76]. Additionally, enhancing thumb-to-finger interfaces with visual cues has proven effective in improving target acquisition of interactive elements. This approach reduces cognitive load and enhances intuitiveness compared to midair gestures [20, 31, 63]. Over time, users may also develop muscle memory through proprioception and tactile feedback, akin to traditional physical keyboards [23].

We utilize these advantages identified in previous work to develop an unimanual text entry technique that supports ergonomic input and in-situ usability. Additionally, we assess the feasibility of continuous input through a WGK application, contributing to advancements in Hand Proximate User Interfaces (HPUI).

2.3 Word Gesture Keyboards

Word-gesture keyboards enable users to input text by tracing continuous gestures over virtual keyboard layouts, predicting words based on the shapes formed by these gestures [85]. This method offers an efficient and user-friendly alternative to traditional tap-typing by leveraging the fluidity of natural hand movements. Gesture typing has been explored using input methods suitable for HMDs, including gestures detected through controllers [40], motion tracking [58, 67], head rotation [83], and gaze [12, 45]. This body of work underscores the general utility of WGKs as viable and efficient text entry solutions for HMDs. While one-handed tap typing on smartphones has shown to be viable and popularly used, it yields on average 29.2 WPM [62], compared to the average WPM of swiping on smartphones at 39.82 WPM [51]. In addition to the potential for higher efficiency with WGKs, it also can potentially alleviate limited resolution of thumb-to-finger interaction [37]. While the visual feedback would further provide improved precision, prior work has noted the imprecision even with such visual feedback [21]. Consequently, we implement a WGK for improved efficiency, best taking advantage of the limited space available for ThumbSwype.

3 Implementation

3.1 Hand tracking and Interaction Detection

We use the Vicon motion tracking system¹ with 13 cameras due to the limitations of current hand-tracking algorithms on HMDs. This setup provided the precision needed for our current investigation. The hand is tracked with a felt glove with IR markers (see 4a). The tracked hand data was streamed to the Unity application to drive the virtual hand the participants would see. We expect improvements in hand tracking in the future to allow directly applying the keyboard we implement without any additional instrumentation of the hand. The VR application built in Unity (2022.3.9f1) displays the keyboard and related content on and around the hand of the participant. We use a Meta Quest 3² for our studies. The motion capture from Vicons was streamed real-time to Unity to show the user's hand pose at any given time. All data for experiments were captured utilizing the Unity Experiments Framework [7]

To detect thumb-to-finger interactions, the open-source package of Hand Proximate User Interfaces (HPUI)³ was used. Similar to prior work [23], the package allows using deformable meshes that are tracked to the hand using Unity's Skinned Mesh Renderer⁴. These surfaces are made interactive by attaching colliders to each vertex of the deformable mesh. Interactions are then detected using raycasts projected from the tip of the thumb. To minimize the effect of the fat finger problem [69], we make the thumb translucent and provide a small sphere cursor that appears on the surface of the mesh the thumb is either making contact with or is hovering above. When the thumb is making contact, the cursor turns green and turns red when hovering. We add this to provide clear feedback to the participant on what the system is doing.

For the consequent studies we used a Meta Quest 3 headset running our application at 72 fps and a render resolution of 5408 x 2912 (i.e. 1.3x the default render resolution). The headset was driven using PCVR through Meta Quest Link. The desktop computer driving the headset had the following specifications :i5-12600K, 3060 (12GB VRAM) and 32 GB Memory.

3.2 Keyboard Interface

The keyboard is made of five interactive surfaces:

¹<https://www.vicon.com/>

²<https://www.meta.com/ca/quest/quest-3/>

³<https://github.com/ovi-lab/HPUI-Core>

⁴Unity manual on Skinned Mesh Renderer

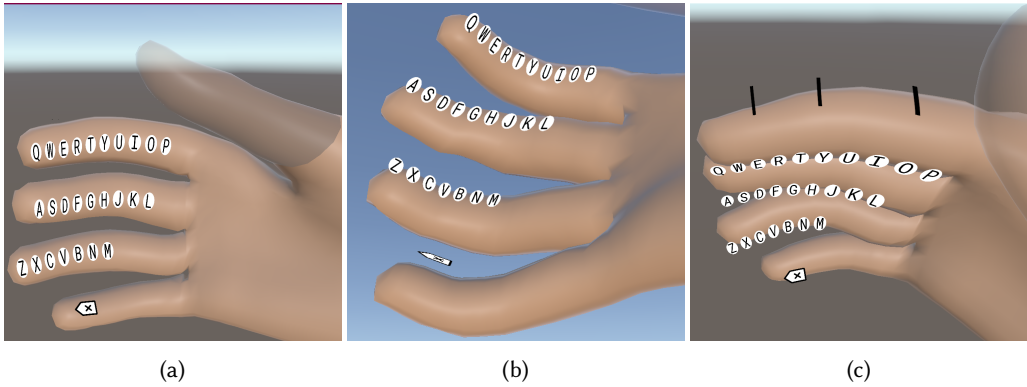


Fig. 2. (A) Displays the ThumbSwype system for text input, with a virtual hand featuring a QWERTY layout mapped onto the fingers. (B) Shows a bottom view of the hand, highlighting individual deformable interactables. (C) depicts the radial side of the index finger with three separation indicators, marking four zones, one for each alternate word suggestion

- Three *keyboard* surfaces on the volar side (or palm facing side) of the index, middle, and ring finger to present the corresponding rows of the keyboard (see 2a)
- One surface for *suggestions*, on the radial side (or the thumb side) of the index finger to make alternate word selections. (see 2c)
- A *backspace* placed on the volar side of the tip of the little finger. (see 2a)

Each row of the QWERTY keyboard layout is displayed on the index, middle, and ring fingers respectively (see 2a). We use the QWERTY layout as it is the most common keyboard layout [16]. Each row starts at the tip of the respective fingers. Since the visuals of the keyboard are placed on deformable meshes as explained in Section 3.1, bending or curling causes the letters to partially stretch or squeeze (see Figure 2b). The user can start typing with ThumbSwype by touching their thumb down to any part of their index middle and ring fingers, and then maintain contact with the fingers as they perform a swipe similar to swipe typing on smartphones. Once the swipe is complete the path of the swipe is used to compute the intended word. We first describe how the gestures are computed here before elaborating how the words are computed from the gestures in subsection 3.3

The deformable surfaces report the point of contact on the surface and only one surface would be in contact at any given time. The gesture is computed by projecting the point of contact on any of the three keyboard surfaces onto a single plane. Rather than utilize the full width of the index, middle and ring fingers, we only use the index surface to determine the width of a key. This is because the top row in the QWERTY layout has the most number of keys. Subsequently, on average, a single key width approximates to 6.31mm, and this is maintained along the middle and ring fingers. We avoid using the proximal phalanges (the innermost volar portions of the fingers) of the middle and ring finger, as prior research on thumb-to-finger interactions have outlined the lack of comfort when performing interactions in those regions [37, 63]. Note that, because the surfaces follow the tracked fingers, based on the length of the participants' fingers, the lengths of the individual surfaces along the fingers would be different. Hence, when projecting the contact points onto a single plane, they are normalized by the length of the surface. This results in the locations of the individual keys of the QWERTY layout on the projected plane to be the same for all participants under all conditions. We refer to this setup as the *keyboard plane* in the remainder of this paper.

This plane is also used to compute the *system template gestures* discussed in subsection 3.3. The sequence of projected contact points on the plane is then used as the path of the swipe gesture. When a swipe gesture on the keyboard surfaces completes, it computes and displays the word as well as four alternate words on a suggestions panel underneath the text box (see figure 6b). The suggestions and backspace surfaces act as regular interactive surfaces. The suggestions surface maps to the four suggestions offered by splitting the radial surface of the index finger to four parts. Based on the area tapped, the corresponding word would swap with the current swiped word. For instance, if the user wanted to substitute their current word “for” with the first suggested (leftmost) word “good”, the user would tap on the leftmost radial portion of their index finger (example in Figure 6). Tapping anywhere on the backspace surface would delete the last word entered.

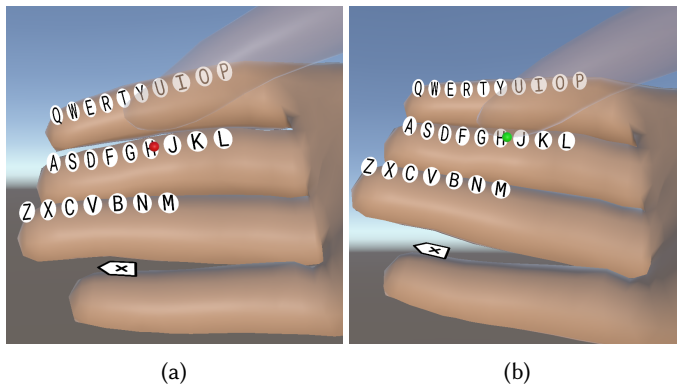


Fig. 3. (A) The cursor turns red when the user’s thumb hovers over a letter. (B) Cursor turns green when the user’s thumb makes contact with the interactable planes.

The detection of interactions using colliders can result in false positives, particularly when the participant moves their thumb over the fingers quickly. For example, when the participant wants to tap the backspace on the little finger, the raycasts from the thumb may accidentally make contact with the surface on the ring finger. To mitigate such instance triggering an input action, an initial time threshold is applied. If an interaction with a surface lasts less than this threshold, the interaction is discarded. Similarly, as the keyboard interface is made of three separate surfaces, when a swipe gesture is performed across the fingers the thumb may momentarily not be in contact with any of the surfaces. Despite it being part of the same gesture, the momentary lapse in contact results in the system perceiving it as separate gestures. To avoid this, we also include an exit time threshold. After the thumb stops making contact with any of the surface (which is when an interaction with any of the surfaces is considered to be complete) if another gesture starts within the exit time threshold, the two gestures are considered to be part of the same gesture. In our studies, we use 0.06 seconds initial time threshold and 0.3 seconds exit time threshold. These values are derived based on implementation tests such that the above issues are mitigated while reducing any perceived lag in the systems’ response. Also note that, as the thumb transitions from one surface to another during a gesture, the projected points on the keyboard plane can be slightly staggered as the contact point is normalized by the length of the surface along the finger and projected to the keyboard plane. But, as we see in the remainder of this paper it does not impact the final results.

3.3 Parsing Gestures

SHARK2 [43] is the most prominent technique for implementing Word Gesture Keyboards (WGKs). ThumbSwype also utilizes the same technique to detect the words. The dictionary used in the ThumbSwype keyboard during the studies was sourced from the Google Web Trillion Word Corpus [60]. From this, we selected 10,000 of the most frequently occurring words. Since our evaluation used the Mackenzie Phrase Set [55], we made sure to include words present in the phrase set that were not in the top 10,000 most frequently occurring words. This selection of words was then filtered for profanity, producing a final list of 10111 words. For every word in this dictionary, a corresponding shape is recorded to represent the word. This is done by using the position of each key on the QWERTY layout relative to the center of the keyboard plane, for all the letters in a word, and then interpolating all the intermediate points. Words where double letters are present (such as "appointment") are treated as a single letter. Every word is thus represented by a collection of p points connecting together to form a gesture, and this acts as our collection of system template gestures. Based on implementation tests, $p=100$ gave enough resolution for all words present in the dictionary.

Once a gesture is completed, it is passed to the gesture decoder to parse. The dictionary is pruned to a subset of relevant words based on the n -closest letters to the starting and ending point of the user's gesture. Implementation tests revealed $n=4$ for closest starting letters and $n=6$ for closest ending letters to be viable. This subset of words was then analyzed for the closest match via shape matching algorithms. Section 4 discusses in detail the evaluation and implementation of the shape matching algorithm. The results of the shape matching algorithm are then integrated with unigram statistics using on Bayesian probabilities [25, 53]:

$$P(w|g; freq) = P(w|g) \cdot P(w|freq)^{-\alpha}$$

Here $P(w|g; freq)$ represents the final probability of the word. $P(w|g)$ is calculated based on the scores reported by the gesture decoding component, which we expand on in section 4. The top k ranked words are then combined with their probability of occurrence based on unigram frequency $P(w|freq)$. α here controls the degree of influence of the language model over the final output. The frequency numbers for the words were sourced from the Google Web Trillion Word Corpus [60]. By limiting the language model to only influence words already identified by the gesture decoding model, we reduce the chances of words with orders of magnitude high frequencies (such as "the") from showing up during invalid scenarios.

4 Preliminary Study: Improving the Accuracy of ThumbSwype

The gestures on ThumbSwype are being performed on a dynamic, non-contiguous surface. Additionally, the raycasts to colliders-based approach for determining the contact point introduces noise to the gesture path. We required a decoding process that would be robust under these conditions. To determine this, we conduct a study where the goal was to select the optimal way of filtering the noise from the user's gesture, and then find a way to match the user's gesture with our system template gestures for the most likely words. We considered two options noise filters: One Euro Filter [9] and Unscented Kalman Filter [77]. Both these filters have well established in literature, with One Euro Filters being used in human tracking research [59, 61], and the unscented Kalman having more industrial applications [2, 5, 57].

We aimed to investigate the effects of a faster but more jittery cursor compared to a slower but smoother cursor. We selected the following parameters to further explore this during the pilot study: (1) A Fast one Euro Filter: $f_{cmin}=0.8$, $\beta=0.35$ (**EF**), (2) A Smooth one Euro Filter: $f_{cmin}=3$, $\beta=0.3$ (**ES**), (3) A Fast Kalman Filter: $Q=0.05$, $R=0.3$ (**KF**), (4) A Smooth Kalman Filter: $Q=0.3$, $R=0.05$ (**KS**), (5) We also analyze the outputs from the **raw** gesture data as a baseline. These parameters were

determined on the basis of initial pilot testing, aiming to examine the trade-offs between latency and noise reduction for these filters.

As for shape matching, we decided to compare between a Mean Euclidean Distance (MED) [72, 78, 83] and Dynamic Time Warping (DTW) [45, 53, 65]. MED compares the average distance between the user's gesture and the system gesture, by resampling the user's gesture to the same number of points as the system template gesture. MED is calculated as follows:

$$S = \frac{1}{N} \sum_{i=1}^N \|U_i - G_i\|$$

This gives a score S based on the average distance between the points for the User's Gesture and a given System Template Gesture,

DTW is an elastic pattern-matching approach that uses a dynamic programming approach to match two time series signals which may be similar in overall shape, but not necessarily over the same timestamps. Given two sequences U (user's gesture) with i points and T (system template gesture) with j points, the similarity score S is given by $D(i, j)$ where

$$D(i, j) = d(U_i, T_j) + \min \begin{cases} D(i-1, j) \\ D(i, j-1) \\ D(i-1, j-1) \end{cases}$$

here $d(U_i, T_j)$ is the cost of aligning points U_i and T_j .

When implementing DTW in multiple dimensions, two approaches can be taken for computing the cost. DTW can be computed per dimension, such that the cost is simply the absolute difference between values on a given dimension

$$d(U, T) = |T_i - U_i|$$

which is then computed per dimension and averaged for a final score. This is referred to as DTW Independent [68]. Alternatively, the cost calculation can be computed as the Euclidean distance between the two points:

$$d(U, T) = \sqrt{\sum_{i=1}^n (T_i - U_i)^2}$$

This is referred to as DTW Dependent [68]. While a multitude of WGG implementations [13, 44, 45, 53] discuss implementing DTW in order to decode the spatial component of the gesture, it is not made clear which variant of DTW is being implemented. Prior research in robotics signal processing has indicated that neither approach is necessarily better than the other, and performance needs to be analyzed on a case by case basis [68]. Consequently, we decided to analyze the performance of both variants for a more thorough exploration.

4.1 Participants

Participants were recruited for this pilot via an internal mailing list. With approval from our university Research Ethics Board, we recruited $n=12$ participants (10 Male, 2 Female, average age 27.0 (sd. 4.06)) for the pilot study. The pilot duration was one hour.

4.2 Procedure

Participants would perform swipe gestures for words randomly sampled from the Mackenzie Phrase set [55]. The words were selected such that all letters of the alphabet were covered, and all possible word lengths within the phrase set were also covered. Note that all participants were shown the same subset of randomly sampled words. Swipes would be parsed through all combinations of

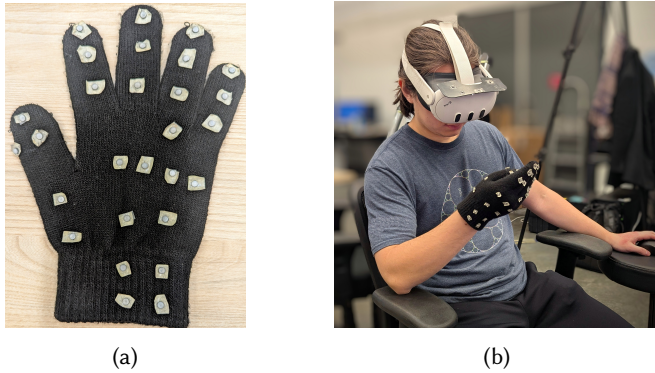


Fig. 4. (A) Felt glove fitted with 28 retro-reflective markers (B) A participant using the ThumbSwype system

noise filters and shape matchers. This results in a score for all words in the dictionary for each condition. The words are then ranked by the score received for each condition. They would progress to the next word if the expected word was in the top 10 predicted words. Otherwise, the participant would see the same word again.

To minimize participant frustration, they would progress to the next word if they failed to perform a gesture that scored the intended word in the top 10 of any condition three times consecutively. Note that the aim of this study is to compare the performance of the filtering and shape matching algorithms. Hence, in addition to the above choices, the language model described in subsection 3.3 is not used here.

Participants were briefed on the functionality of the system, and instructed to wear the Quest 3 VR headset and the motion capture glove. Each participant was given practice trials to familiarize themselves with the system. Participants were prompted with one word at a time, and instructed to "swipe as accurately as possible, ensuring the cursor passed over all the relevant letters present in the word".

12 participants performed swipes for 10 blocks with 10 words per block for a total of $12 \text{ (participants)} \times 10 \text{ (blocks)} \times 10 \text{ (words)} = 1200$ swipes. Note that this number is slightly lower than the actual number of swipes performed (1412), on account of repetitions from invalid swipes (271), which were not analyzed. This resulted in a total of 1141 valid swipes analyzed.

4.3 Measures

This pilot utilized a within-subject design with two independent variables: *Filter* with five levels (Raw, EF, ES, KF, KS) and *Shape Matcher* with three levels (MED, DTW Independent, DTW Dependent) for a total of $5 \times 3 = 15$ conditions. To avoid fatiguing participants, all combinations of swipes and filters were run simultaneously in the background. As mentioned in Section 4.2, if any filter and shape matching combination yielded the expected word within the top 10 predicted words, the system would move to the next trial. For the dependent variable, we analyzed the *top-N accuracy*, where $N \in \{1..6\}$. A participant's top-N accuracy for a given condition is the percentage of swipes where the expected word's score appears in the first N results for that swipe. For example, the top-3 accuracy of DTW Dependent and ES (One Euro Smooth filter) would indicate how often the target word was present within the top-3 word suggestions for that filter and shape matcher combination.

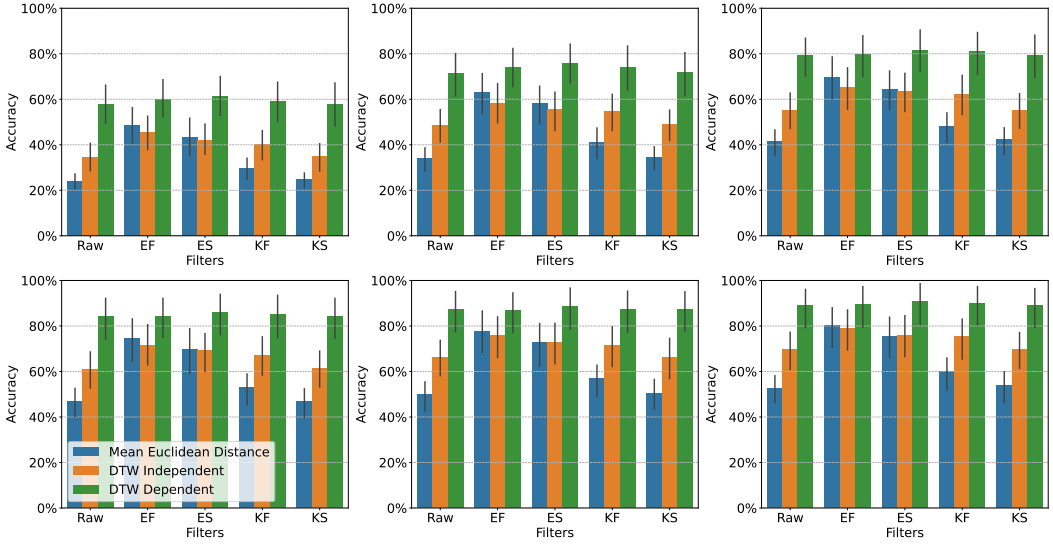


Fig. 5. Top $k = 6$ Rank Accuracy of different noise filtering and shape matching algorithm combinations on ThumbSwipe

4.4 Results

The summary of the results can be seen in Figure 5. For all N considered, the data met the constraints of RM-ANOVA, and for each $N \in \{1..6\}$, we conducted RM-ANOVA on the top- N accuracy with the filters and shape matchers as factors. The RM-ANOVA results are summarized in Table 1. For all N both factors had a main effect on the top- N accuracy. There were also interaction effects between the factors for all N .

To explore the interaction effect, for each value of N we conducted post-hoc comparisons with Bonferroni correction. For all filters, across all values of N , pairwise comparisons show significant differences between Mean Euclidean Distance (MED) and DTW Dependent, as well as between DTW Independent and DTW Dependent shape matchers. When one Euro filters were used (ES & EF), there was no significant difference between MED and DTW Independent shape matchers. Also, for all values of N , when either MED or DTW Independent shape matchers were being used, the one Euro filters had the highest mean accuracy values. When the MED shape matchers were used, we observed significant differences between all filters except between the Kalman filter variants and the raw data. In addition, we see the Kalman filters performing relatively poorly. However, this only implies that this filter is ill-suited for the task at hand. Meanwhile, when DTW Dependent was used as a matcher, there were fewer significant differences between filters, suggesting the choice of filter does not significantly impact the overall performance of DTW Dependent. It is also worth noting that for either filter, there is mostly no significant difference between the smooth and fast variants. This implies there is little to no trade-off between smooth and fast filters. However, in all cases, the smooth variants had the higher mean accuracy. Finally, the DTW Dependent with the ES filter (one Euro smooth) had the highest mean accuracy in all comparisons, suggesting this combination to be the most preferred for our technique. These patterns can also be observed in the Figure 5.

⁵For the interpretation of effect sizes, Cohen's guideline was followed [11]

Table 1. Summary of the RM-ANOVA results and effect size (η_G^2) from the pilot study and related statistics for each top-N where $N \in \{1..6\}$.

	Filter		Shape matcher		Interaction (Filter:Shape matcher)	
	$F(4, 40)$	eff. size	$F(2, 20)$	eff. size	$F(8, 80)$	eff. size
top-1	69.74 ●	0.18 ○	66.86 ●	0.50 ●	14.20 ●	0.10 ○
top-2	75.81 ●	0.20 ○	98.19 ●	0.51 ●	27.99 ●	0.15 ○
top-3	69.19 ●	0.18 ○	113.06 ●	0.50 ●	32.18 ●	0.14 ○
top-4	79.32 ●	0.18 ○	120.54 ●	0.49 ○	30.23 ●	0.15 ○
top-5	66.25 ●	0.18 ○	103.54 ●	0.49 ○	41.37 ●	0.16 ○
top-6	68.77 ●	0.18 ○	94.45 ●	0.49 ○	36.79 ●	0.16 ○

●: $p < 0.001$, ●: large effect⁵, ○: moderate effect⁵, ○: small effect⁵

While our optimization experiments highlight some interesting relations between the different noise filters and shape matching techniques, we do not claim that this would be valid under other input sensing mechanisms such as using capacitive fabrics [80]. Rather we wish to highlight that such an optimization process should be empirically conducted when implementing WGKs with novel techniques to improve the overall performance of the keyboard.

5 Evaluation of ThumbSwype

We conducted a user study to evaluate ThumbSwype. Our main focus was to investigate overall viability of the technique. This includes objective metrics such as WPM, error rates, interaction statistics and subjective feedback. Additionally, the relation between user's average one-handed smartphone swipe performance and ThumbSwype was of interest as well.

5.1 Participants

With approval from our university Research Ethics Board, we recruited 19 participants from our university to participate in our study. Three participants could not complete the study. Of these, two failed because the motion-tracking systems' calibration failed, and one participant was unable to reach their little finger even though we only use the tip of the little finger in our study (for the backspace). Note that prior work also has encountered similar participants [20]. Of the 16 participants who completed the study (15 male, 1 female) the average age of participants was 22.5 years (sd. 2.31), and of the 16 participants, 15 were right-handed and one was ambidextrous. While our current implementation only accounts for right-hand dominant users, they reported being comfortable using their smartphone with their right hand, and so proceeded as normal with the study. Self reported prior experience with Virtual and Augmented Reality headsets averaged at 2.88 (sd.1.93) out of 7, while prior experience with swipe typing or gesture typing on smartphones was 2.75 (sd. 2.77) out of 7. The study duration was one hour, and all participants received a \$20 gift card as compensation for their time.

5.2 Procedure

After signing a consent form, the participants were briefed on how the keyboard works and the procedure of the study. They then completed a demographic questionnaire. To ensure familiarity with swipe typing, and to serve as a baseline, participants performed a 30-second test on the open source web-app MonkeyType.com⁶. Participants were instructed to perform this test one-handed

⁶<https://monkeytype.com>

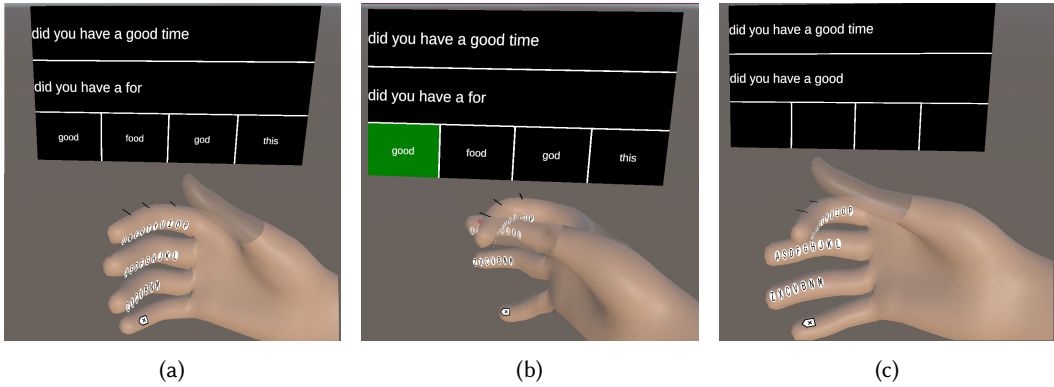


Fig. 6. (A) The prompter and text box interface for the main evaluation study. The topmost box shows the phrase to be transcribed, the middle box acts as the user’s input space, and the 4 boxes on the bottom provide alternate suggestions for the current swiped word (B) User notices the current word “for” does not match the intended word “good”, and selects correct word from alternate suggestions (C) The incorrect word is replaced with correct word

and use swipe typing. This was done on a Google Pixel 7 Pro device with the default Android keyboard with no typing history to prevent automated personalization. We recorded the results for further analysis, similar to prior work [53, 74]. We avoid further priming with swipe typing as we wanted to analyze the typical usage performance with minimal training overhead. They are then asked to take a seat in the motion capture volume and wear the Quest 3 headset along with the tracking glove on their right hand. Once they wear the headset, the study application is launched. Each trial of the study consisted of phrases randomly selected from the Mackenzie Phrase Set [55], which the participants were required to transcribe. Note that the same random subset was presented to all participants in order to ensure consistency, but their order was shuffled. A floating interface anchored to the hand presented the phrase and the text-input space, along with the alternate suggestions underneath the user’s text box (see 6b). A block consisted of 5 such trials, and the entire experiment consisted of 8 blocks. The participants are then given a practice session consisting of two blocks (with different phrases from the main experiment) to grasp the system’s functionality. After the participant completes the practice trials, they proceed to the study. Each block in the study consists of 5 trials after which the participant is encouraged to take a rest. A mouse is provided by their left side to move to the next trial. The participants can move from one trial to the other by double-clicking the right-button on the mouse with their left hand. Participants are instructed to be as fast and accurate as possible, and type as though they were “sending text messages to friends and colleagues”, and to “ensure the prompt and their input matched before proceeding to the next trial”. After completing the 8 blocks of the main study, participants are then required to fill a questionnaire asking for their overall comfort with the system, and a standardized System Usability Scale (SUS) [6], along with asking for their overall impressions of the system and feedback.

5.3 Measures

Because our goal was to benchmark baseline text-entry performance, we did not vary any interface or task conditions for this evaluation study. The following measures were analyzed to understand the performance of ThumbSwype.

Similar to other WGK implementations [53, 58], we calculated Words Per Minute (WPM) and accuracy, along with the distribution of phrase transcription actions performed. WPM is given by

$$\text{WPM} = \frac{|T| - 1}{S} \times 60 \times \frac{1}{5}$$

where $|T|$ denotes the length of the phrase transcribed, and S denotes the time difference from the first touch to the last touch. $|T| - 1$ is used, as the time to transcribe is calculated from the first recorded touch (accounting for the time to perform visual search for the first character [56]). Accuracy is given by Word Error Rate (WER) which is computed on the basis of Minimum String Distance (MSD) adapted to calculate on a word level as Minimum Word Distance (MWD) [3, 13, 86].

$$\text{WER} = \frac{\text{MWD}(U, T)}{|T|} \times 100\%$$

Note that this increases the error rate relative to a character level comparison, since even if a single character is incorrect in a word (for example, "this" and "thus"), the entire word is treated as incorrect. The length of the string is the number of words present in either the transcribed string or the prompt, whichever is larger.

We translate the actions performed by participants during the transcription typing task into four intents: (1) *Match*, where the user performs a swipe, and follows by performing another swipe, thus implicitly indicating they have matched the current word with the intended word, (2) *Select*, where the user performs an alternate word selection, (3) *Undo*, where the user performs a backspace right after swiping, indicating none of the presented outputs were valid and undoing the swipe, and finally (4) *Delete*, where the user notices an error present in the transcribed sentence earlier, and performs multiple backspace operations to rectify it. Finally, the average gesture duration and inter-gesture duration were also tracked.

Regarding subjective feedback, participants had to fill out a standard System Usability Scale [52], and indicate their overall comfort with the system on a 7-point Likert scale. They were also interviewed post the experiment for their general impressions and feedback for the technique.

5.4 Results

A total of 640 phrases ($16(\text{participants}) \times 8(\text{blocks}) \times 5(\text{phrases})$) were transcribed during the study. We first elaborate on the objective metrics, then report on subjective feedback and finally explore correlations between participants one-handed swiping performance on smartphones and ThumbSwipe.

5.4.1 Words per Minute. Participants averaged 14.5 WPM (sd. 6.17) across all blocks. Averaging the highest and lowest WPM per block across all participants netted 20.28 WPM (sd. 0.82) and 8.99 (sd. 1.06) respectively. As the data did not meet the normality constraint, we ran a Friedman Test on average WPM with the blocks. The Friedman Test did not reveal any significance for blocks in average WPM across the 8 blocks ($p = 0.06$, $\chi^2 = 13.19$). Figure 7a highlights the block-wise WPM performance averaged across all participants.

5.4.2 Accuracy. WER averaged at 1.01% (sd. 4.46) over the entire experiment. As the data did not meet the normality constraint, we ran a Friedman Test to check for significance between blocks. The Friedman Test did not reveal any significance for blocks in WER across the 8 blocks ($p = 0.79$, $\chi^2 = 3.9$). Figure 7b highlights the block-wise WER averaged across all participants.

5.4.3 Interaction Statistics. Table 2 documents the percentage of each action executed during the study, and Figure 8a shows the block-wise distribution of these actions. At 64.11%, users performed Matches more than any other category. This validates the system's accuracy from the analysis

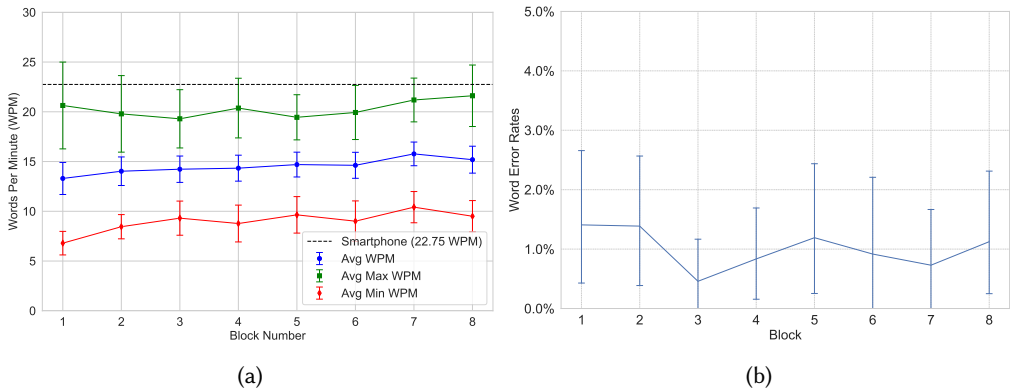


Fig. 7. Typing metrics: (A) Typing Performance Envelope with 95% confidence intervals, (B) Percentage Word Error Rates with 95% confidence intervals

Table 2. Frequency of the various phrase transcription operations

Intents	Match	Select				Undo	Delete
		1st	2nd	3rd	4th		
Count	3015	371	199	78	74	736	230
Percentage	64.11%	7.89%	4.23%	1.66%	1.57%	15.65%	4.89%

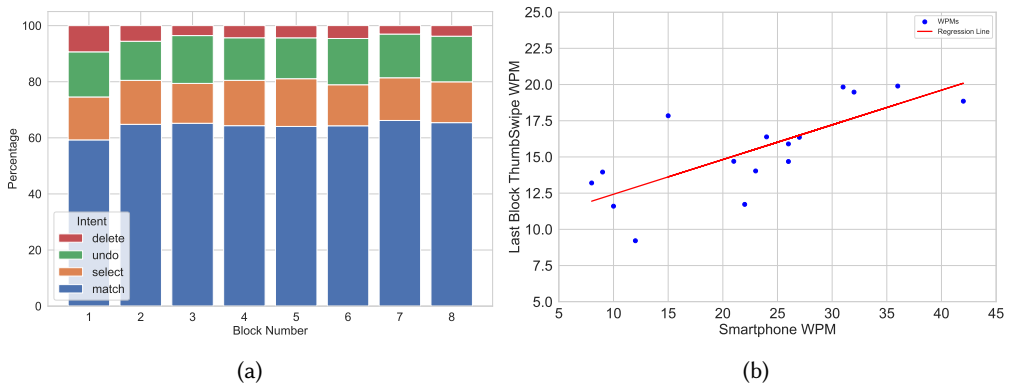


Fig. 8. (A) Stacked Bar Chart of Intents by block (B) Regression analysis between Smartphone and Thumb-Swipe last block performance

performed in Section 4.4. On average, participants performed 1.15 Undo operations per trial, with 15.65% of all actions being Undo operations. As for alternate suggestions, 15.35% of operations involved selecting an alternate word. Of these, the first suggestion was the highest, accounting for half of all alternate word selections. Finally, participants performed the Delete operation 4.89% of the times.

The mean time for a swipe gesture was recorded at 2.395 seconds (sd. 1.12), with intervals between gestures lasting on average 1.243 seconds (sd. 0.44). This does not account for the time required to read the next word, recognize mistakes or pick from alternate suggestions.

5.4.4 Subjective Feedback. The usage of SUS as a metric was utilized to gauge the overall user experience with the keyboard. The overall SUS Score was recorded at 76.3, and a comfort rating of 5.44 (sd. 0.96) on a 7 point Likert scale with a median comfort rating of 6. Prior work [52] discusses possible interpretations of the scores, and a score of 76.3 would indicate "Good" to "Excellent" usability. However, since we do not compare with other similar techniques, this figure should be interpreted with caution.

Participants appreciated the interface of the technique being analogous to one-handed smartphone usage, with multiple participants (PID 1, 3, 8, 10, 11, 15) commenting about "initially being a bit lost", but "getting the hang of it very quickly". Participants also noted the value of the technique in encumbered scenarios, with PID 16 stating "Once I get into the groove, this feels easier than smartphone typing... I can see this being very useful if I have my other hand holding a bag or something else." PID 2 and 14 noted "At some point it felt as natural as typing on my phone", and PID 11 stated "I feel like this would be really nice to do in public rather than wave my arms around", indicating the potential social acceptability of the technique.

When asked for feedback or drawbacks of the system, few participants indicated "It sometimes feels too sensitive, and other times not so much", highlighting the imprecise manner of touch detection. PID 3 and 8 noted "The alternate word selection is finicky" when intending to substitute their current output, but accidentally triggering a new word swipe on account of both interaction planes being very close to each other. This caused significant friction in having to end the new accidental swipe, perform two delete operations to remove the accidental new word and the previous incorrect word, and then swipe the previous word once again. PID 8 indicated "It would have been nice if the suggestions for each of the words were preserved when I deleted my current word", however such functionality could have confounding effects on the overall speed of the system and cause confusion in disambiguating intent of an action. PID 8 and 15 also expressed interest in having cursor controls, indicating their interest in using ThumbSwype as a regular means of text input.

5.4.5 Smartphone vs ThumbSwype Performance. Similar to prior work [53, 74], we had collected typing performance on via smartphone typing for a short period as a baseline metric. We performed a regression analysis to examine the relationship between participants' MonkeyType scores on their smartphones and their average words per minute (WPM) on ThumbSwype in the last (8th) block. The analysis yielded an adjusted R^2 of 0.52, indicating that approximately 50.2% of the variance in the WPM for the last block can be explained by the MonkeyType scores (see Figure 8b). Additionally, Pearson's correlation test reported a correlation coefficient of 0.745, suggesting a strong positive relationship between the two variables. For the average WPM across all blocks, adjusted R^2 was 0.5021, with 0.732 for Pearson's correlation test.

6 Discussions

6.1 Contextualizing the performance of ThumbSwype

Herein we would like to compare our results with the works most closely related to ours. As with any text entry method, results vary due to differences in procedures, systems, and the technique's focus, and are difficult to directly compare. Multiple text-input methods run longitudinal studies to establish learning effect and see the performance increase over time. However, our main goal was to validate the interaction technique for text entry. In the future, we intend to further explore how to optimize and study how the text entry performance evolves with a longitudinal study.

Techniques such as Vulture [58] and RotoSwype [29], which maintain QWERTY layouts, show an initial performance between 5-12 WPM, which is significantly lower than our system. RingGesture [66] achieves an impressive 27.3 WPM, however they report that the biggest contributor to

their speed improvement was integrating a contextual LSTM to assist with better predictions. The Naive Correction model used is similar to ours, and performs at 17.6 WPM, similar to ThumbSwipe, indicating that ThumbSwipe could also potentially benefit from a significant speed increase by incorporating the novel ScoreFusion technique described by Shen et al. [66].

When contrasted against novel techniques such as PinchText [38] and H4VR [24], there is a meaningful difference in WPM (12.7 and 4.8 respectively) compared to ThumbSwipe. This can presumably be attributed to the lack of prior familiarity with the technique. Finally, microgesture techniques such as TipText [82] and DRG Keyboard [53] achieved around 12 WPM. While this is highly impressive given the size of the keyboards implemented in both these techniques, performance has to be traded for robustness and higher need for precision on the part of the user, leading to overall slower speeds.

It's worth noting however, that the technique is arguably not performing at the highest possible speed yet. Currently, 31% of the interactions constitute of text editing operations rather than text input (alternate word selection and undo operations). The limited input space of the technique, along with fat finger accuracy issues [36, 75] both contribute to the reduced accuracy of the user. Purely machine learning based approaches [1, 66] have shown to be viable solutions for swipe decoding, especially for unique gesture based techniques. Collecting and training on user swipes might prove to be more effective at reducing the requirement for performing editing operations.

Additionally, about 35% of the time is not spent on performing gestures (mean inter-gesture duration). This is spent on analyzing the output, and inspecting if rectifications need to be made. A part of this could be attributed to unfamiliarity with the technique, where muscle memory over extended use would reduce the time to undo or pick a specific alternate word. This could also be modeled by simulated expert usage studies [48, 70]. The text interface could additionally be improved to reduce the cognitive load required to perform the transcription task, by closely mimicking a smartphone like design, whereby the text panels would directly extend from above the hand rather than float above it. This also implies free-form text entry might be faster, such as casual messaging or browsing the internet.

6.2 Easy transferability of smartphone typing experience

As noted by Section 5.4.5, there is a clear correlation between existing one-handed swipe experience on smartphone and performance on ThumbSwipe. Additionally, as discussed in Section 5.4.1, there was no significant difference in the performance over the duration of the experiment. While many keyboards experience a learning effect, either via longitudinal studies [29, 58, 66], or via single sessions [53, 74, 78], for ThumbSwipe participants acquired a high initial WPM from the very first block, and maintained that pace till the end of the study. This indicates the mapping that participants are easily able to establish with their prior experience on an interface closely mimics the smartphone counterpart. As noted by Gustafson et al. [30], developing new techniques which resemble frequently used existing interfaces causes the user to inadvertently perform "transfer learning", and quickly familiarize themselves with what is a completely new input medium otherwise. Further research investigating the long term performance improvements might find the gap between one-handed smartphone text input and ThumbSwipe shrink, as users adapt to account for the underlying differences between the two systems.

6.3 Exploring Additional Affordances

In our current implementation, ThumbSwipe does not explicitly allow for character level input, and subsequently can not add new Out of Vocabulary (OoV) words to the dictionary. However, with the assistive cursor, enabling such functionality would be trivial, by simply treating each letter as a gesture of points present at the letter's position itself (as is currently handled for the words

"a" and "I"). This would consequently support adding Out of Vocabulary words to the dictionary, which would then be accessible via swipes. Additional considerations would have to be taken for integrating character level input with gesture type, for instance the placement of the space key. Taking inspiration from more novel one-handed techniques such as FingerText [48] might be of interest, where the user may perform actions such as space via tapping on a fingernail instead.

While we did not directly examine eyes-free usage, thumb-to-finger interactions and hand-proximate user interfaces inherently support eyes-free input through proprioception [23, 30]. As discussed in Section 5.4.4, users who reported daily swipe usage on smartphones anecdotally noted performing swipes for shorter words without looking at the cursor, relying instead on muscle memory and their mental keyboard model. This presents an exciting direction for future research, as midair techniques lack comparable proprioceptive feedback and eyes-free accuracy.

Similarly, ThumbSwipe being one-handed implicitly affords multitasking with the other free hand. Explorations into performing texting with the dominant hand and a secondary task with the non-dominant hand (or vice versa) might reveal interesting findings, especially combining it with eyes-free input. Additionally, integration with alternate input modalities like gaze and speech could be an interesting future exploration.

7 Limitations and Future Work

While the technique is applicable for HMDs which can visualize a keyboard on the user's fingers, we test the technique primarily in VR with a virtual replica of the hand, with a translucent thumb. Extension of this technique to Augmented Reality would require modifying the visuals of the keyboard such that the portion of the keyboard underneath the thumb would be partially faded to indicate depth. Moving the technique purely to motion tracking via HMDs would also require fairly advanced hand pose estimation. While recent techniques have explored robust hand tracking for gestures [32] and flat surface typing [73], ThumbSwipe makes consistent use of hand postures which are self occluding, which is notably difficult to estimate via egocentric cameras. Approaches utilizing EMG bands [46], bio-acoustics [47], electrical sensing [41], egocentric hand tracking [32], or more likely some combination of the above with sensor fusion would ideally eliminate the need for marker based tracking solutions such as the Vicons. This would also mean the technique would no longer require felt gloves, however, we do not expect any performance differences.

Given the relatively small space for performing the swipes, there is a certain level of dexterity expected from the user. Degradation of hand-movements with age might hinder such motions. An alternate form of this technique could utilize both hands, where a larger keyboard would be placed on the non-dominant hand, similar to PalmType [79]. Swiping on this larger surface would require lesser precision from the user, and the decoder could perform with higher accuracy as well.

The assumption we are validating with the evaluation study is the transferability of prior swipe typing experience, and we do not compare with another text input technique in VR. The smartphone swipe performance acts as our baseline. Future work would contrast ThumbSwipe with related one-handed implementations such as RingGesture [66] or smartphone akin techniques like STAR [74] for a more comprehensive analysis of objective and subjective metrics.

Finally, the utilization of a 10,000 word dictionary for this technique was in accordance with prior WGK implementations for research [53, 58]. Extending this list to a larger corpus of 40,000 words [8] may increase the number of errors by virtue of overlapping gesture shapes [43]. This might necessitate a better gesture decoder, or optimizing the layout of the keyboard [70] while retaining the general form of the QWERTY layout [4] to preserve familiarity.

8 Conclusion

ThumbSwipe introduces an efficient and comfortable one-handed text entry technique for head-mounted displays, leveraging users' familiarity with smartphone typing. By overlaying a QWERTY keyboard on the fingers and utilizing thumb-to-finger swipes with tactile feedback, ThumbSwipe enables quick and precise input without extensive training. Our studies demonstrate that users can achieve an average typing speed of 14.52 WPM with minimal practice, reaching peak performances close to their smartphone speeds. ThumbSwipe shows great potential as an effective text input method for head mounted displays.

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